



Extended Loss: Incorporating Long Context into Training Models when using Short Audio Frames

Quang Minh Dinh^{*1}, Hoda Rezaee Kaviani², Mehrdad Hosseinzadeh², Yuanhao Yu²

¹Simon Fraser University, Canada

²Noah's Ark Lab, Huawei Technologies Canada Co., Ltd., Canada

qmd@sfu.ca, hoda.rezaee.kaviani@huawei.com, mehrdad.hosseinzadeh@huawei.com,
yuanhao.yu@huawei.com

Abstract

Recently deep learning solutions have been successfully applied to many signal processing tasks including acoustic echo cancellation (AEC). Most existing works focus on architecture design and ignore practical issues such as the effect of frame length on the performance of end-to-end AEC models. In real-time applications, frame length can be as small as 10ms. Since the observed context is very limited during training, it results in boundary discontinuities (glitches) in the final output. While using long frames or post-processing can help, it adds extra delay which may not be desirable depending on the application. In this paper, we investigate the practical issue of handling short frames for AEC, and propose an efficient remedy. By keeping the long context information in each batch and using it during loss calculation, we compensate for the short frames. Our solution is model-agnostic and does not affect the inference time.

Index Terms: Echo cancellation, signal processing, neural networks, speech enhancement, real-time processing

1. Introduction

In recent years, the popularity and demand for teleconferencing systems have increased significantly. The existence of acoustic echos is one of the major sources of poor speech quality ratings in voice and video calls. Acoustic Echo Cancellation (AEC) aims to recover clean near-end audio, given the noisy and far-end signals. Early methods for this task employ different digital signal processing (DSP) techniques; however, these methods do not perform well when model assumptions are violated, e.g., non-linearities in the processing chain, or fast time-varying acoustic conditions [1]. With the advent of deep learning techniques, many supervised acoustic echo cancellation (AEC) techniques have been proposed and outperformed their traditional counterparts. These models either directly operate on time [2, 3, 4] or frequency domain [5, 6, 7].

Amongst the most recent methods, CRUSE [5], Align-CRUSE [6] and DeepVQE [7] belong to a family of residual CNN auto-encoder with a Gated Recurrent Unit (GRU) bottleneck models that perform especially well on different speech enhancement tasks, including echo cancellation. Another architecture successfully applied to AEC tasks is Conformer [8], which combines the power of CNNs and Transformers. [9] and [10] are conformer-based models trained with an ASR loss and an auxiliary loss, and are highly effective on the AEC task.

In the design of AEC solutions, it is common to divide the input signal into multiple fixed-length frames and process them independently by the network. During training, regardless of their domain or architecture, all of these methods cal-

culate loss function on a single frame. Processed frames are later concatenated to form the final output. One major problem with this approach is that the lack of contextual information between adjacent frames during training results in boundary discontinuity (glitches) in the output. In the context of speech enhancement, [11] discusses the same issue and proposes a way of handling short frames for a specific waveform-based speech enhancement model by adding a recurrent neural network (RNN) module to explicitly model the relationship between adjacent frames. Other solutions like applying the overlap-add method [12] or predicting the current frame given the surrounding context [13] have been proposed which result in additional delay.

In this paper, we study the influence of frame length on the performance of end-to-end deep learning-based AEC systems and propose a solution to handle the problem of lack of context when training models on short audio frames. Instead of relying on post-processing, or the existence of neighboring frames during inference, we keep the long context information in each batch during training and use it during loss calculation. The existence of this information allows us to compensate for short audio frames during training. For the experiments we chose two of the most recent AEC architectures, CRUSE [5] and DeepVQE [7], which is the state-of-the-art on AEC task, and implemented our Conformer based AEC model. To the best of our knowledge, this practical issue has not been addressed in designing AEC systems. We show that our proposed solution is architecture-independent and can work with various loss functions. The contributions of our work can be summarized as follows:

- We study the practical issue of handling short frames in AEC tasks and propose an efficient solution to address this problem.
- Our method is model-agnostic. To support this claim, we explore a few of the most recent architectures (including SOTA) and loss functions and compare the results.
- Proposed method achieves better results without adding extra complexity to the model, or affecting the inference time.

2. Proposed Method

In an acoustic echo cancellation system, a far-end signal is digitally transmitted to the other side, played through a loudspeaker at near-end (depending on the scenario, a near-end signal can also add to the mix simultaneously. The mixed signal can contain the background noise as well). The mixed signal at the near-end is then transmitted back to the far-end side leading to the far-end hearing the "echo" of their own signal.

There are three different possible scenarios for the mixed signal to happen: 1) far-end single talk (FEST): only the far-end speaker is talking and the near-end is quiet. 2) near-end single

^{*}Work done during internship at Noah's Ark Lab, Huawei Canada

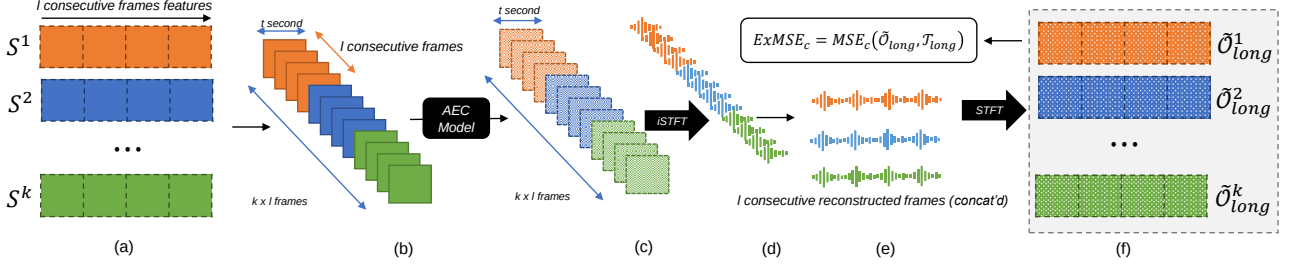


Figure 1: Overview of the training pipeline. (a) First k audio files are randomly selected from the training set, and STFT features for l consecutive short frames are extracted $S^1 \dots S^k$. (b) Each batch is formed in a way that consecutive frames appear sequentially, and $b = k \times l$. (c) The output of the AEC model is the enhanced audio per short frame. (d) Inverse STFT is applied on each short frame separately to obtain their reconstructed waveforms. (e) l consecutive waveforms that belong to the same audio are concatenated together to form a long waveform. (f) STFT is applied on the long sample which is used for loss calculation. In the regular non-extended setup, loss calculation is performed at stage (c) on predicted short audio frames.

talk (NEST): the far-end signal is quiet and only the near-end microphone is talking. Ideally in this scenario, no suppression should happen. 3) double-talk scenario (DT): when far-end and near-end are talking over each other. This is arguably the hardest scenario as the far-end signal should be suppressed as much as possible while the near-end signal must be kept intact.

2.1. Input Features

We first sample a batch of size b from of the far-end and microphone signals at a pre-defined sample rate. Let $F \in \mathbb{R}^{b \times 1 \times n}$ and $N \in \mathbb{R}^{b \times 1 \times n}$ where n is the number of samples to be the far-end and microphone signals, respectively. We sample all audio signals in a mono fashion, hence F and N both have 1 channel. Next, the raw sampled audios are converted to a spectrum feature space. Following other works in the literature [7, 6] we also use Short-time Fourier Transform (STFT) to project the sample audio waveforms to the feature space:

$$\mathcal{F} = [\mathcal{F}_r, \mathcal{F}_c] = STFT(F) \in \mathbb{R}^{b \times t \times f \times 2} \quad (1)$$

$$\mathcal{N} = [\mathcal{N}_r, \mathcal{N}_c] = STFT(N) \in \mathbb{R}^{b \times t \times f \times 2} \quad (2)$$

where t and f are the time and frequency axes for the Short-time Fourier Transform, respectively. Note that although the STFT operation generates complex output, we only keep the real part of the transformation as model input.

2.2. Model Architecture

Since our solution is model-agnostic, we use a variety of models with different numbers of parameters (from 0.6M to 12M) to showcase the effectiveness of our proposed loss function. The de facto standard model design choice for the AEC task is mask-based architecture. In this approach, the network takes the near-end and far-end signals as input, either through separate branches or stacked together, and predicts a mask. The predicted mask is then applied to the real and complex part of $\mathcal{N}$ via simple multiplication to get the echo-free output, $\mathcal{O}$:

$$\mathcal{M} = \theta(\mathcal{F}_r, \mathcal{N}_r) \in [0, 1]^{b \times t \times f} \quad (3)$$

$$\mathcal{O} = \mathcal{M} \cdot \mathcal{N} \in \mathbb{R}^{b \times t \times f \times 2} \quad (4)$$

where $\theta(\cdot, \cdot)$ denotes the AEC model parameters. The intuition for the mask-based approaches is that (ideally) the mask modifies the component of the near-end features that are responsible for the existing echo in the signal. After applying the mask on

the microphone features, a corresponding inverse transformation (e.g. inverse STFT) generates the waveform for the clean signal.

Specifically, we use CRUSE [14], DeepVQE [7], and Conformer [8] models as our backbone AEC model. For each model, we aim to modify as little as possible to make them suitable for the task of acoustic echo cancellation. Since DeepVQE was originally designed for the AEC task, we didn't modify the architecture. As for CRUSE and Conformer models, since they were originally proposed for the task of speech enhancement and noise cancellation, modification is needed to adapt them for the task of AEC. Therefore, we stack the near- and far-end signals together and feed them as a single input to the model.

2.3. Loss Function

The main contribution of this work is our proposed Extended-Loss function. We argue that assembling a batch informed by a long context and using this information during loss calculation, plays a crucial role in the overall performance of the model. We start by defining the standard loss function and elaborate on how we extend it.

Let $\mathcal{O} \in \mathbb{R}^{b \times t \times f \times 2}$ and $\mathcal{T} \in \mathbb{R}^{b \times t \times f \times 2}$ be the output and target (i.e. enhanced mic. and clean ground-truth mic.), spectrum complex features, respectively. The standard ComplexMSE [15] loss is a spectral-based loss function that is calculated on the real and imaginary parts of output and target spectrum features for each sample, and then averaged over all the samples existing in the batch:

$$MSE_c = \frac{1}{b} \sum_{i=1}^b \lambda * |\tilde{\mathcal{O}}_c^i - \mathcal{T}_c^i|^2 + (1 - \lambda) * |\tilde{\mathcal{O}}_r^i - \mathcal{T}_r^i|^2 \quad (5)$$

In this setup, MSE operates on each sample individually, and the final average over the MSE of the samples in a batch will be the same, regardless of the order of them. In this scenario, the maximum context available to the network is as long as the sample itself, and each batch is usually completely shuffled during training. This setup can be very challenging for training models with short frame length due to lack of long context.

Prior work [6, 16] show that applying inverse STFT on the output followed by a STFT again, before applying MSE loss function can generally improve the audio quality. It is also a common practice to use a combination of different loss functions to improve the overall quality of the audio. However, in all

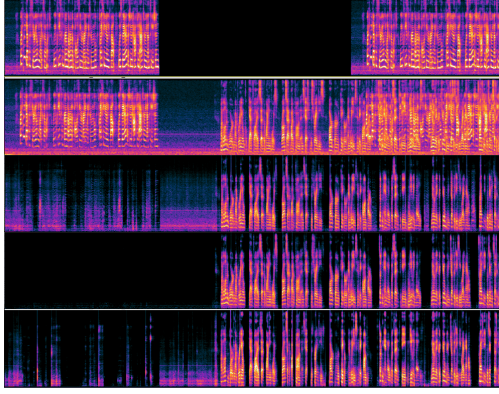


Figure 2: From top to bottom: 1) Far-end 2) Double-talk microphone signal 3) Enhanced by a model trained on 368ms samples without extended loss 4) Enhanced by the same model, trained on 368ms samples with extended loss 5) Enhanced by the same model trained on 10s audio, and used for inference on 368ms audios.

of these solutions, the maximum context available to the model is still as long as the frame processed by the network.

To overcome this issue, we propose a novel batch formation and loss computation technique, that allows the model to use long context information without changing the sample size. In non-extended setups, each batch of size b , is formed by sampling short random frames from different audio files. In our setup, when sampling a frame from a random audio, we make sure to include its next l consecutive non-overlapping frames as well, and preserve their order when forming the batch. Random sampling is performed k times until we reach batch size $b = k \times l$. The batch is now processed by the network without the need for any extra modification to the architecture of the model. The output of the model is the predicted mask $\mathcal{M} \in [0, 1]^{k \cdot l \times t \times f}$ and is multiplied by the near-end STFT features to obtain $\mathcal{O} \in \mathbb{R}^{k \cdot l \times t \times f \times 2}$.

For loss calculation, instead of treating each sample individually, we first apply iSTFT to the predicted short frames to obtain $k \times l$ reconstructed waveforms. Then we concatenate l consecutive frames that belong to the same audio, to get k long waveforms. Note that up to this point no MSE or ComplexMSE is calculated. Finally, STFT is applied on the long waveforms: $\tilde{\mathcal{O}}_{long} = STFT(\mathcal{O}_{long}^i)$ and Extended ComplexMSE loss function is calculated:

$$\text{ExMSE}_c = \text{MSE}_c(\tilde{\mathcal{O}}_{long}, \mathcal{T}_{long}) \quad (6)$$

where $\tilde{\mathcal{O}}_{long}, \mathcal{T}_{long}$ are the output and target complex spectrum features for k long samples. This procedure is shown in Figure 1. In the non-extended setup, loss is calculated at stage (c) of Figure 1, on the predicted short audio frames. Regardless of the choice of loss function, the common non-extended setup treats each short frame individually, while our method takes advantage of longer context that we preserved in the batch during loss calculation.

3. Experiments

3.1. Datasets

Our training dataset is synthesized on-the-fly using DNS Challenge [17] dataset with random parameters for each sample

(e.g., signal-to-noise ratio, distortion, signal-to-echo ratio). To generate training samples, we randomly select clean speech audios, level normalize them, and concatenate them to the desired duration. For both near and far-end, the clean speeches are separated by 500ms to 2s silence segments. Noise segments are either sliced or repeated to the target duration. Echo is synthesized by convolving the far-end speech signal with a random small room RIR, before being mixed with the near-end speech signal with a signal-to-echo ratio (SER) between 5 and 25. Noise is then added to the echo mixture with a signal-to-noise ratio (SNR) uniformly selected between 5 and 10. We re-scale the final mixture to a random target level between -35 and -15 dBFS.

We use the globally aligned version of the real recordings from the blind test set in the AEC Challenge 2023 [1] to evaluate our models, as delay is not the main concern of this work.

3.2. Evaluation Metrics

For both DT and FEST scenarios, we report the AECMOS score [18] which is a neural network-based perceptual speech quality metric that is highly correlated with the human mean-opinion-score (MOS). For FEST, we also report ERLE, which is only appropriate if measured in a quiet room without background noises. For NEST, we report the AECMOS degradation score [18], (which is correlated to the call echo degradation quality), DNSMOS P.835 signal (SIG), background (BAK), and overall (OVRL) scores. We also employ the AECMOS degradation score for DT in addition to the AECMOS echo.

3.3. Implementation Details

All audios are sampled at 16kHz. For feature generation, we use a squared root Hann window of length 32ms, a hop length of 16ms, and a discrete Fourier transform length of 512. Following [5], for all CRUSE experiments we use a squared root Hann window of length 20ms, a hop length of 10ms and a discrete Fourier transform length of 320. We use the AdamW optimizer with $LR = 1.5 \cdot 10^{-4}$ and a weight decay of $5 \cdot 10^{-6}$. Experiments for extended setup are with 5 frames unless stated otherwise.

3.4. Ablation Study

To study the effect of our extended setup, we conducted experiments with different numbers of frames. Results are presented in Table 1, along with an example of enhanced DT spectrogram shown in Figure 2. The model used for all experiments in this table is a conformer with fixed architecture, and the data pipelines for all experiments have the same configuration*.

We chose a long (10s) and short (368ms) duration for both training and inference. It can be seen that the single-frame model trained and tested on longer duration (row 4), generally performs well on all scenarios. To demonstrate the importance of context, we trained and tested the same model on short audios (row 1). As expected, performance dropped significantly across all metrics. Row 5 shows the inference results on shorter audios using the model trained on long frames. We observe that changing the duration of the inference contributes to notable degradation, especially in FEST and NEST. The major drop in ERLE shows this model does not perform well on removing echo. Although the AECMOS echo for DT stayed close to row 4, the audio quality was significantly degraded.

*Sample of audio results for subjective comparison available at <https://quangminhding.github.io/ExtendedLoss/>

Num frames	Train duration per frame	Inference duration	DT		FEST		NEST			
			AECMOS _e	AECMOS _d	AECMOS _e	ERLE	AECMOS _d	SIG	BAK	OVRL
1	368ms	368ms	4.09	2.77	4.08	15.43	3.14	3.02	3.82	2.63
3	368ms	368ms	4.34	2.97	4.19	30.19	3.59	3.14	3.95	2.8
5	368ms	368ms	4.44	2.99	4.35	38.31	3.5	3.13	3.97	2.79
1	10s	10s	4.6	3.71	4.21	25.89	3.93	3.27	3.98	2.93
1	10s	368ms	4.57	2.41	4.19	14.64	2.56	2.73	3.88	2.43

Table 1: Ablation study results for our Conformer models on the AEC Challenge 2023 blind test set.

Method	Frame duration	No. params	DT		FEST		NEST			
			AECMOS _e	AECMOS _d	AECMOS _e	ERLE	AECMOS _d	SIG	BAK	OVRL
CRUSE	370ms	0.65M	3.43	3.1	4.02	20.64	3.29	3.01	3.64	2.5
x5 frames	370ms	0.65M	4	3.11	3.96	20.72	3.61	3.03	3.76	2.59
DeepVQE-S	368ms	0.67M	3.62	2.13	2.97	22.98	2.44	2.6	3.82	2.26
x5 frames	368ms	0.67M	3.97	2.32	3.17	28.67	2.57	2.6	3.87	2.28
DeepVQE-S	10s	0.67M	4.12	3.06	3.83	32.29	3.6	3.04	3.94	2.71
x5 frames	10s	0.67M	4.22	3.17	3.65	40.35	3.52	3.06	3.97	2.72
DeepVQE	368ms	8M	4.11	2.51	3.56	21.6	2.7	2.65	3.89	2.34
x5 frames	368ms	8M	4.4	2.84	3.68	30.62	2.91	2.74	3.93	2.42
Conformer	368ms	12M	4.37	2.05	4.3	32.8	2.6	2.75	3.91	2.37
x5 frames	368ms	12M	4.45	2.61	4.51	56.98	3.23	2.99	3.97	2.64
Conformer	10s	12M	4.61	3.45	4.76	137.79	3.68	3.19	3.96	2.84

Table 2: Ablation study results on the AEC Challenge 2023 blind test set.

To show the effectiveness of our method, we trained the same model on short audios with two extended setups: 3-frames (row 2), and 5-frames (row 3). Both of these extended setups outperform the single-frame model (row 1) across all metrics. One interesting observation is that our 5-frame extended model outperforms row 4 in FEST. It is worth noting that the difference in the context used for training is more than 5 times for the model in row 4 compared with row 3 (10s vs 5x368ms), and the difference in context at inference is more than 27 times (10s vs 368ms).

In Figure 2 we show samples of enhanced spectrograms on the same DT signal, with 3 different models. All models have the same architecture, and the inference duration is short (368ms) for all models. The first and second rows (from top to bottom) correspond to the far-end and microphone signals, respectively. The 3rd row is enhanced by a model trained on short audios without extended loss, and row 4 is the same architecture, trained on short audios with our proposed extended loss. Between the two, our method performs significantly better in removing echo and noise. This is specifically more visible in the first half of the signal. The last row, is the same architecture, trained on long 10s audios, while inference is 368ms. This model performs better than row 3, but still has some left-over echos and noise (more visible in the first half of the signal), compared with our results in row 4. This is consistent with our observation that changing the inference time, affects the quality of final audio.

To show our solution is architecture-independent, we trained other models (CRUSE, DeepVQE, and DeepVQE-S), with and without our extended setup. The results are shown in Table 2. We observe that all of the extended setups outperform their corresponding single-frame version. For DeepVQE-S, we experimented on both long and short frames. Using the extended setup improves the results for both scenarios. For all experiments, we used complex compressed MSE loss, ex-

cept for Conformers, where we used a weighted combination of complex compressed MSE, asymmetric loss, and SNR loss on the active areas of the speech. It can be seen that the combination of Conformer with the new loss already outperforms many of the other experiments. What is more interesting is that our extended setup on this model boosts the results even further. The last row of this table shows our best model, performing exceedingly well on all metrics, especially ERLE for FEST, and significantly outperforming other SOTA architectures.

4. Discussion and Conclusions

In this paper, we investigated the practical issue of handling short frames and its effect on the performance of deep learning-based AEC systems. While using longer frames or adding a post-processing step can be an option in some cases, this may not be a solution for real-time applications due to latency and complexity. However, the lack of context associated with a short frame hurts the training of the model. We proposed a solution to handle the lack of context during training without adding extra complexity or delay to inference by utilizing intra-batch context information in the loss function during training. This allows us to compensate for the short length of each frame during training. Our solution is model-agnostic and does not affect on inference time.

5. References

- [1] R. Cutler, A. Saabas, T. Parnamaa, M. Purin, E. Indenbom, N.-C. Ristea, J. Gužvin, H. Gamper, S. Braun, and R. Aichner, "Icassp 2023 acoustic echo cancellation challenge," 2023.
- [2] S.-W. Fu, T.-W. Wang, Y. Tsao, X. Lu, and H. Kawai, "End-to-end waveform utterance enhancement for direct evaluation metrics optimization by fully convolutional neural networks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 26, no. 9, pp. 1570–1584, 2018.

- [3] D. Rethage, J. Pons, and X. Serra, "A wavenet for speech denoising," *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5069–5073, 2017. [Online]. Available: <https://api.semanticscholar.org/CorpusID:24512955>
- [4] A. Pandey and D. Wang, "A new framework for cnn-based speech enhancement in the time domain," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 7, pp. 1179–1188, 2019.
- [5] S. Braun, H. Gamper, C. K. A. Reddy, and I. J. Tashev, "Towards efficient models for real-time deep noise suppression," *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 656–660, 2021.
- [6] E. Indenbom, N.-C. Ristea, A. Saabas, T. Pärnamaa, and J. Guzin, "Deep model with built-in self-attention alignment for acoustic echo cancellation," *ArXiv*, vol. abs/2208.11308, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:251765344>
- [7] E. Indenbom, N.-C. Ristea, A. Saabas, T. Pärnamaa, J. Guzin, and R. Cutler, "Deepvqe: Real time deep voice quality enhancement for joint acoustic echo cancellation, noise suppression and dereverberation," in *INTERSPEECH*, 2023, pp. 3819–3823.
- [8] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, "Conformer: Convolution-augmented transformer for speech recognition," 2020.
- [9] S. Panchapagesan, A. Narayanan, T. Z. Shabestary, S. Shao, N. Howard, A. Park, J. Walker, and A. Gruenstein, "A conformer-based waveform-domain neural acoustic echo canceller optimized for asr accuracy," 2022.
- [10] T. O'Malley, A. Narayanan, Q. Wang, A. Park, J. Walker, and N. Howard, "A conformer-based asr frontend for joint acoustic echo cancellation, speech enhancement and speech separation," *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pp. 304–311, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:244463264>
- [11] N. Abuhajar, T. Sun, Z. Wang, S. Gong, C. D. Smith, X. Wang, L. Xu, and J. Liu, "Network compression and frame stitching for efficient and robust speech enhancement," in *NAECON 2021 - IEEE National Aerospace and Electronics Conference*, 2021, pp. 269–276.
- [12] W. Zhong-Qiu and W. Shinji, "Improving frame-online neural speech enhancement with overlapped-frame prediction," *IEEE Signal Processing Letters*, vol. 29, pp. 1422–1426, 2022.
- [13] P. Chuan, T. Lan, M. Li, S. Li, and Q. Liu, "Frame-level speech enhancement based on Wasserstein GAN," in *Eleventh International Conference on Signal Processing Systems*, ser. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, K. Mao, Ed., vol. 11384, Dec. 2019, p. 113840G.
- [14] R. Cutler, A. Saabas, T. Pärnamaa, M. Loide, S. Sootla, M. Purin, H. Gamper, S. Braun, K. V. Sørensen, R. Aichner, and S. Srinivasan, "Interspeech 2021 acoustic echo cancellation challenge," in *Interspeech*, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:239694056>
- [15] A. Ephrat, I. Mosseri, O. Lang, T. Dekel, K. Wilson, A. Hassidim, W. T. Freeman, and M. Rubinstein, "Looking to listen at the cocktail party: a speaker-independent audio-visual model for speech separation," *ACM Transactions on Graphics (TOG)*, vol. 37, no. 4, pp. 1–11, 2018.
- [16] S. Wisdom, J. R. Hershey, K. Wilson, J. Thorpe, M. Chinen, B. Patton, and R. A. Saurous, "Differentiable consistency constraints for improved deep speech enhancement," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 900–904.
- [17] H. Dubey, A. Aazami, V. Gopal, B. Naderi, S. Braun, R. Cutler, H. Gamper, M. Golestaneh, and R. Aichner, "Icassp 2023 deep noise suppression challenge," in *ICASSP*, 2023.
- [18] R. Cutler, B. Naderi, M. Loide, S. Sootla, and A. Saabas, "Crowd-sourcing approach for subjective evaluation of echo impairment," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 406–410.